

High-Dimensional Regression: $p > n$

Every chapter so far has assumed, openly or quietly, that there are more observations than coefficients. Once $p > n$ the columns of $\mathbf{X}$ cannot be linearly independent, least squares fits the data exactly whatever they are, and no individual coefficient is estimable. This chapter asks what can still be done. The first answer is to keep least squares and pick the shortest of its many solutions. Its risk can be computed exactly, and it produces the “double descent” curve. The second answer is to assume that only a few coefficients matter and to use the lasso. Here a restricted form of the eigenvalue conditions of Part II replaces full rank, and short non-asymptotic proofs show that the lasso predicts and estimates almost as well as an oracle that knew which coefficients were zero. Selecting exactly the right variables needs a stronger condition, which can fail even when prediction is good. The chapter ends with inference. Intervals computed after selection are wrong, and a one-step correction of the lasso restores approximately normal estimates of single coefficients.

What you need. [Chapter 1](#) (the singular value decomposition, Theorem 1.8.1, and the Moore–Penrose inverse, Definition 1.3.2), [Chapter 6](#) (projections, minimum-norm least squares, Proposition 6.10.1, and the Frisch–Waugh–Lovell theorem, Theorem 6.6.2), [Chapter 8](#) (estimable functions, Theorem 8.2.2), [Chapter 10](#) (least squares through the SVD, Theorem 10.4.1), [Chapter 13](#) (the union bound, Theorem 13.4.1) and [Chapter 14](#) (random regressors and the inverse Wishart mean, Lemma 14.2.4). From this part: the lasso, its optimality conditions and soft thresholding ([Chapter 27](#), Definition 27.4.1 and Theorem 27.4.2) and ridge regression (Theorem 27.2.2). The random-design sections work with the moments and the normal theory of [Chapter 2](#) and [Chapter 3](#) (Theorem 2.2.1, Theorem 3.3.1, Theorem 3.2.1), the last of these for the tail bounds of [Section 28.3](#). Selection bias and cross-validation ([Chapter 29](#)) are referred to but not needed.

28.-1.1 Roadmap

- [Minimum-norm interpolation](#)
- [Restricted eigenvalue conditions](#)
- [Prediction and estimation bounds for the lasso](#)
- [Support recovery](#)
- [Post-selection and debiased inference](#)
- [Summary and notes](#)